LLMs Are Entering THE RF DESIGN LAB

Large language models (LLMs) are reshaping how engineers approach antenna design, simulation, and testing. Here is what that looks like in practice.



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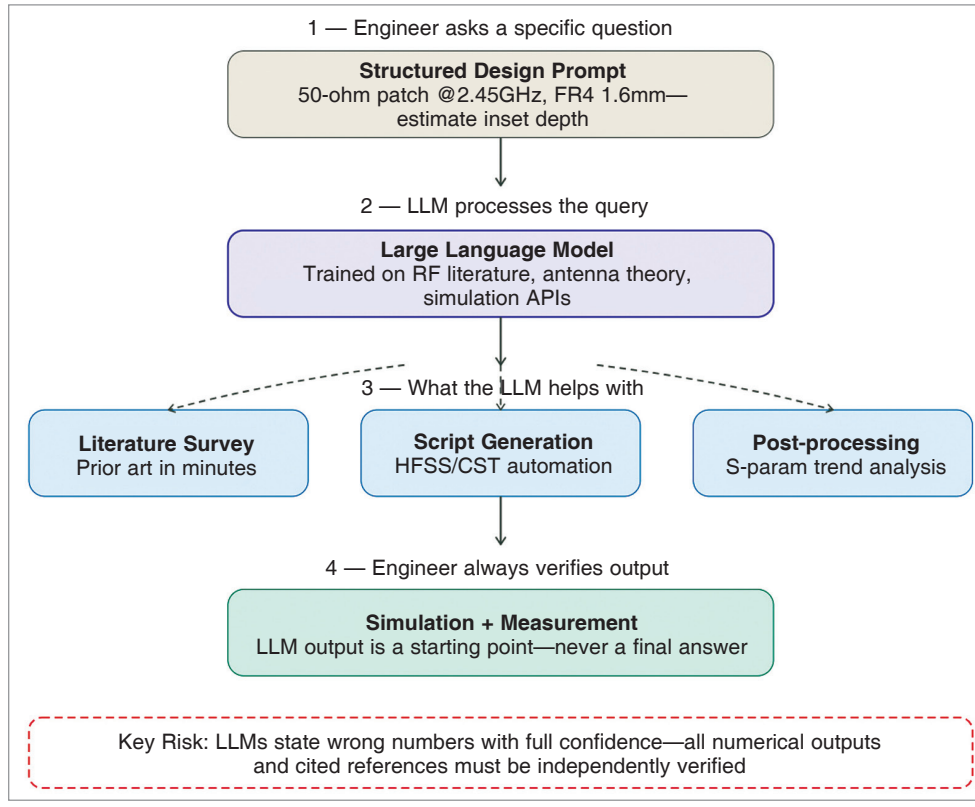
A colleague recently described asking an AI assistant to help debug a microstrip patch antenna simulation that had been producing inexplicably elevated cross-polarisation levels. He expected the tool to fail. Instead, it walked him through a systematic analysis of substrate permittivity tolerances, feed point asymmetries, and simulation mesh artefacts, correctly identifying the issue as likely being a combination of the second and third factors. It took eleven minutes. The same process would have taken most of an afternoon if done manually.

This is not a story about AI replacing RF engineers. It is a story about a tool that, until very recently, had little useful to offer antenna designers, beginning to earn a place in the RF design workflow. Large language models (LLMs) have arrived in the engineering world, and while they are far from infallible, their practical utility in specific RF design tasks is increasingly becoming difficult to ignore.

What LLMs are actually good at in RF work

The honest answer requires resisting two temptations: overselling the technology and dismissing it outright.

LLMs are not good at performing precise numerical calculations. They are not reliable substitutes for electromagnetic simulation software. They cannot replace the physical intuition that comes from



LLMs in the RF design lab